In-wheel gearbox for Formula Student car

SDU-VIKINGS

Indhold

Der blev ikke fundet nogen elementer til indholdsfortegnelsen.

Purpose

Design an in-wheel gearbox for the Viking X. To ensure that the mass is kept low, optimization tools and FEM tools are to be used in the designing of the gearbox.

Introduction

The SDU-Vikings is a team of students at the University of Southern Denmark, who compete in the Formula Student competition. In this competition, the students design and manufacture a car with an open cockpit and uncovered wheels within a ruleset and compete against teams from other universities across the world in competitions taking place in multiple countries, eight in Europe. The team must defend their engineering decisions to judges, present a business case for the car and also present a detailed report of the costs of manufacturing the prototype car.

After these static events, which are worth 325 points in total, the car has to pass technical scrutineering before competing in the dynamic events which are skid pad for competing on purely lateral acceleration, 75m acceleration for competing on longitudinal acceleration, sprint/autocross for competing on handling and cold-car performance and finally a 22 km endurance race in which points are given for total time and also for fuel economy.

Since the cars are designed, built and driven by students, safety is a big concern and among the safety measures is that the autocross and endurance tracks are narrow cone-tracks with many corners and chicanes designed to hold the average speeds within 48-57 km/h and the top speeds below 105 km/h. These low average speeds means that the cornering, braking and acceleration performance is much more influential to the lap times than top speeds but unlike actual motorsport, there are no minimum weight for this competition, which leads to very light and agile cars being developed.

The SDU-Vikings are to build the tenth car, the Viking X, which will be an electrical car with a mass of less than 200 kg and a centralized weight of around 45% of the weight on the front wheels. It is decided that this car will be powered by the AMK Racing Kit, which consists of four electrical motors and inverters and with this, the car will have four-wheel drive with outboard motors in all wheels. Having four-wheel drive with independent motors offers a series of potential advantages for an electrical car as the load on each wheel can be controlled independently so that the available traction is used more effectively and that the regenerative braking can be much more powerful and thus allow a downsizing of the batteries.

The motors from AMK has a maximum torque of 21 Nm and a maximum rpm of 20.000, so a reduction gearbox is needed to get a reasonable top speed and a higher torque at the wheel. This project is about designing this gearbox.

Requirements

The gearbox is required to connect to the AMK motor output shaft and to a wheel with a 4x100mm bolt pattern. The gearbox should sit inside the wheel while still leaving space for the brake disc, brake caliper and suspension.

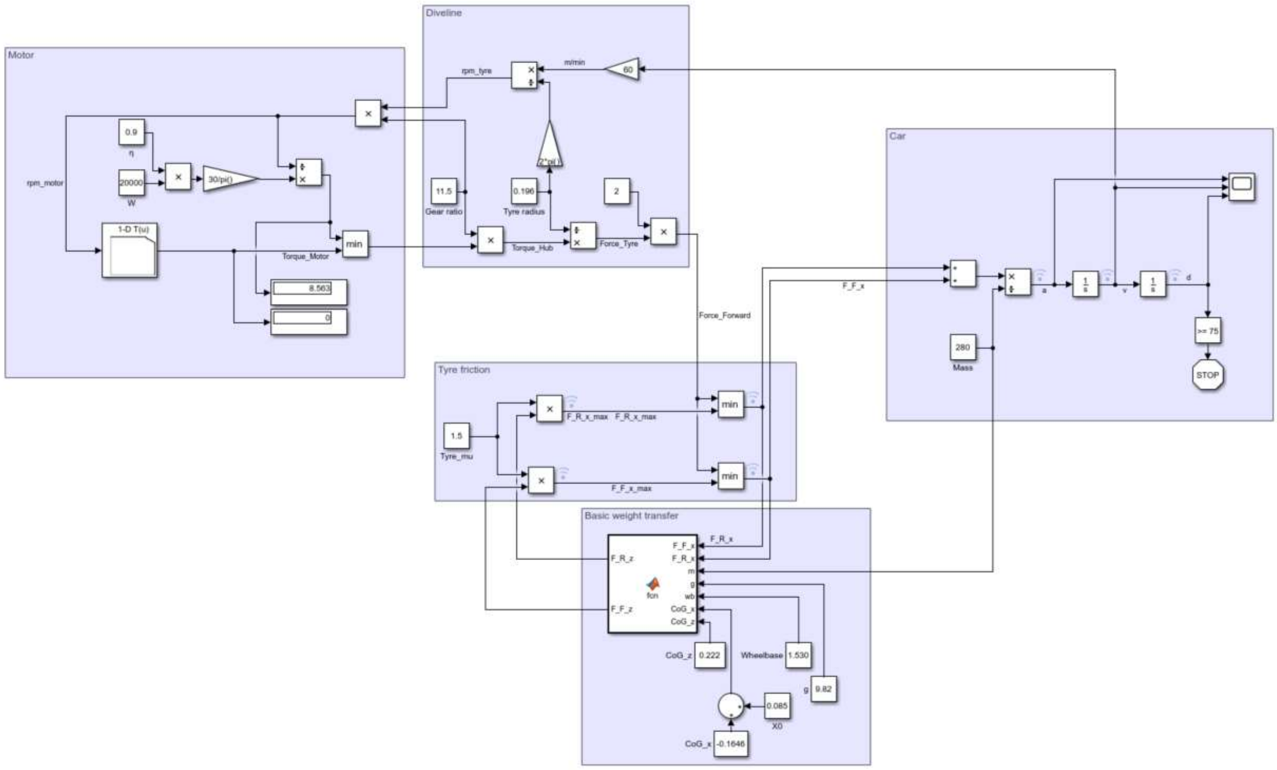
To reduce complexity and costs, only one gearbox is to be designed, for use in all corners of the car.

Determining the gearing ratio[1]

The gear ratio affects both the speed and the torque of a transmission and by this, it can limit both acceleration and top speed. The torque at the output shaft will be that at the input multiplied by the gear ratio and the speed at the output shaft will be that of the input multiplied by the inverse of the gear ratio.

One preliminary calculation could be to decide on a top speed and calculate the required ratio to reach that speed. With a rolling radius of 196,7 mm (measured on Viking IX and referenced to TTC data) a ratio of 10,6 would result in a top speed of 140 km/h and a ratio of 11,4 would result in a top speed of 130 km/h. However, as formula student cars are rarely driven at high speeds, choosing the gearing ratio based solely on top speed is not a good engineering decision and thus would possibly cost points in both the design event and the dynamic events.

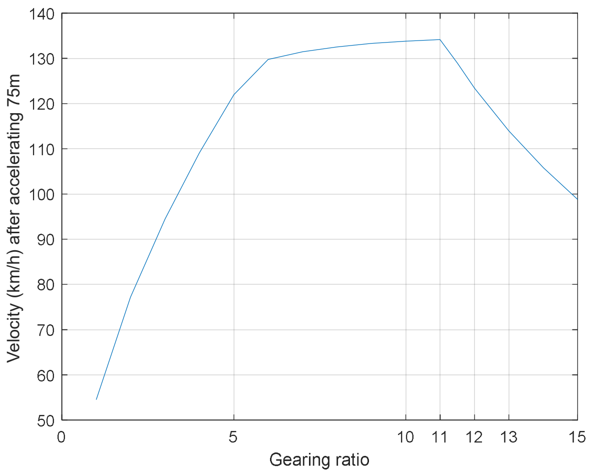
For a slightly more detailed calculation to choose our gearing ratio from, we make a rough simulation of the acceleration event, using a Simulink program developed by Kevin Grau. A perfect traction control and stiff tires are assumed and both rotational inertia and aerodynamic drag are neglected. The tractive force is limited by the friction available on each wheel by assuming a constant coefficient of friction. The output torque of the motors are limited by the motor torque curve and by a maximum of 20 kW per wheel (total of 80 kW as per rule EV 2.2.1) with an efficiency of 90%.

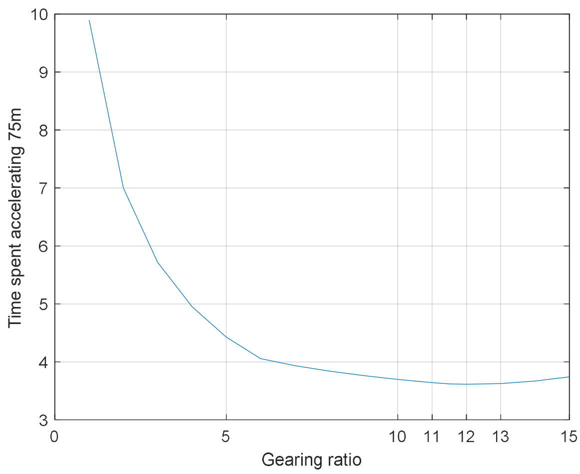


This simulation is run at multiple gearing ratios, assuming a total mass of 280 kg and a coefficient of friction of 1,5 and it is found that above a gearing ratio of 11, the motor will hit the top speed but that the lowest time is at a gearing ratio of 12. The difference in time between a ratio of 11 and 12 are however only 27 ms and due to the coarseness of this simulation, it may only be concluded that the optimum gearing ratio is

[1] This section contains work done by students aside from the author of this paper.

probably between 11 and 12. The actually optimum gearratio may be different due to the influence of mass, rotational inertia and losses.





An even more detailed analysis that would better show the compromises at stake would be a laptime simulator so that the gearing ratio could be chosen to optimize the points across all dynamic events but since a laptime simulator is much more complex, it will not be considered for this gearbox.

Gear type

Gears may be either spur, helical, herringbone (double-helical) or circular. In racecars, spur gears are most often used as these has a few percent less losses than the others. If the drivetrain losses are increased by 1%, then the time spent accelerating 75m in the simulation above is increased by 16 ms, which is the same effect as adding 5 kg to the car in the same simulation. It is unlikely that the benefit of not using spur gears is a weight savings of more than 5 kg to the car and thus, spur gears will be used due to the lower drivetrain losses.

Gear arrangement

Many different arrangements of gears can provide the wanted gear ratio, so we will quickly discuss the pros and cons before settling on an arrangement.

Among the limiting factors of the size of any gearset is the minimum amount of teeth that can be on a pinion before undercutting of the tooth will happen. For a standard tooth with a pressure angle of 20 degrees, undercutting will happen when there are less than 17 teeth but this number can be reduced by applying a profile shift [1]. When profile shifting, the base circle of the involute profile stays the same but the root circle and the tip circle are moved to a larger diameter when the profile shift is positive or to a lower diameter if the profile shift is negative. When a positive profile shift is employed, the minimum amount of teeth reduces to 11 teeth at a profile shift factor between a profile shift of 0,3 and 0,35. The upper limit of profile shift of few gear teeth is set by a requirement that the thickness at the tip should be 0,4 times the module for a hardened gear to prevent the brittle tip from breaking off.

Aside from preventing undercutting, adding a positive profile shift will also make the root of the tooth thicker as thus stronger. When one gear has been given a profile shift, the mating gear must either be given an equal profile shift of the opposite sign or have the distance between the axis of the mating gears moved. For now, the gears will be made in V-zero pairs, meaning that the sum of profile shift in a pair is zero and the axis distance is the same as for a gearpair with no profile shifted gears.

Parallel axis gearset

The most common type of gear arrangement is the parallel axis gearset where one gear sit on an input axis and one gear sit on an output axis parallel to each other. This arrangement is very simple and effective but has the drawbacks that it is not very compact as the size of the casing has to be $(1+i)$ times the diameter of the smaller gear and thus because of the size, it requires more material to make sufficiently stiff as the gears will attempt to push each other apart. The reduction ratio is $i = \frac{z_2}{z_1}$ and the typical efficiency is 98% according to [2].

By making the gearset two-stage, it can be made more compact but still heavier as the load on the second stage gears are increased. The reduction ratio is then $i = \frac{z_2}{z_1} \cdot \frac{z_4}{z_3}$ and the typical efficiency is 97%.

Simple planetary gearbox

A planetary gearbox is a subgroup of epicyclic gearings and in this type, a central gear called the sun is used as an input while one or more planets are in contact with both the sun and a stationary outer ring gear which has internal teeth. The planets are held by a planet carrier that can rotate around the sun and is taken as the output. The ratio is calculated $i = \frac{z_r + z_s}{z_s}$ so if the sun has 11 teeth and a ratio of $i = 11$ is wanted, the ring would have 110 and the planets would have 49 teeth each. This makes this configuration more compact than the parallel axis gearset and also since the multiple engagements transmit the load, the load on each engagement is less and thus the module or the width of the gears can be less than with a parallel axis gearset. Furthermore, under ideal conditions, the load on the sun and ring are evenly distributed and there are no net radial load that causes deflections which concentrates the load on parts of the gear face.

The main drawback of the planetary gearbox is that the structure is more complex and thus more expensive and that it has more bearings and tooth contacts which means that the efficiency is typically 97% as the two-stage parallel axis gearbox.

Simple star gearbox

The star gearbox is a different type of epicyclic geartrain where the planet carrier is held stationary and the ring is taken as an output member. This configuration is slightly less compact than the planetary gearbox as its reduction ratio is $i = \frac{z_r}{z_s}$ but it allows for very significant weight savings if the wheel is redesigned at the same time as the gearbox as this allows for centerless wheels to be used, where the ring gear connects directly to the "barrel" of the wheel.

Since the design of the wheel is not within the scope of this project, this configuration is not considered further despite its advantages.

Stepped-planet compound planetary gearbox

The compound planetary gearbox is a more complex configuration in which one or more planets are used to make a gearing between the sun and the ring gear and with that enable configurations that is not possible with the simple version. The gear ratio of the compound planetary gearbox is

$$i = \frac{z_r z_{p1}}{z_s z_{p2}} + 1$$

The most compact compound planetary gearbox is the stepped-planet compound planetary gearbox in which each axis has two planets coupled rotationally together such that the bigger planet engages the sun and the smaller planet engages the ring gear. This allows the ring gear to be much lower diameter and thus reduce the diameter that the gearbox uses.

The main drawbacks of the compound planetary gearbox are much of the same as the simple planetary gearbox but even moreso since it is more complex and has more bearings. Also, the compound planetary gearbox is more prone to misalignments due to the forces of gear engagement than the simple planetary gearbox. A typical efficiency of a compound planetary gearbox is not stated in [2] and since estimating the efficiency of such a complex mechanism which can have many very different configurations is very difficult, it will be assumed to be approximately the same as the simple planetary gearbox, possibly slightly lower.

The parallel axis and simple planetary gearbox are expected to be so big in outer dimensions that they interfere with the preliminary suspension pickup points and the brake system as they can be placed inside the 10 inch rim. For reasons of compactness, the stepped-planet compound planetary gearbox is chosen for further analysis.

Geometric constraints

All of the epicyclic configurations has a lot of geometrical constraints in order to be assembled and the compound configuration has even more, which will be discussed in this chapter.

In [2] it is found that to be able to assemble, the following two equations has to be fulfilled.

$$D_r = D_s + D_{p1} + D_{p2}$$

And

$$\frac{z_{p1} z_r + z_{p2} z_s}{n_{planets}} \; must\ equal\ an\ integer$$

However, to ensure that the planets are evenly spaced and that they can be made identical with the same phase angle between the two planets that share axis, the second of these constraints need to be extended. An extended version which ensures with identical planets at equidistant spacing is found in a paper [3], by Petru A. Simionescu.

$$\frac{z_{p1} z_r + z_{p2} z_s}{n_{planets}} = A_1 \cdot z_{p2} + A_2 \cdot z_{p1}$$

Where A_1 and A_2 are integers. With D_3 being the greatest common denominator of z_{p1} and z_{p2}, Simionescu states that it is sufficient to search for the integers in the interval $0 < A_1 \leq \frac{z_{p1}}{D_3}$ and $0 < A_2 \leq \frac{z_{p2}}{D_3}$ but this is found to limit our possible configurations by 25% over if the integers are searched for in an interval that is 2000 times larger. All of the solutions found in this very large interval has however also been found if A_1 is being searched for in the interval $0 < A_1 \leq 2 \cdot \frac{z_{p1}}{D_3}$ and A_2 then is simply solved for in the constraint equation without enforcing it to be in an interval.

To ensure that there is enough space for the bigger planets to not interfere with each other another assembly condition is found in [4]:

$$z_{p1} + 2 < (z_s + z_{p1}) \cdot \sin\left(\frac{\pi}{n_{planets}}\right)$$

For the engagement between the planet and the ring gear to function, involute interference and trochoid interference must be avoided. [4]
Trochoid interference is when the addendum of the planet contacts the dedendum of the ring gear but since this happens when the number of teeth of the external and internal gears are near each other, this is assumed not to be an issue.
Involute interference is when the dedendum of the external gear contacts the addendum of the internal gear and happens when the number of teeth of the external gear is low and thus, it is of concern for the engagement between the small planet and the ring. This is avoided when

$$\frac{z_{p2}}{z_r} \geq 1 - \frac{\tan(\alpha_{ar})}{\tan(\alpha_w)}$$

With α_{ar} being the pressure angle at the addendum of the internal gear tooth

$$\alpha_{ar} = \cos^{-1}\left(\frac{d_{br}}{d_{ar}}\right)$$

And α_w being the working pressure angle of the gear pair

$$\alpha_w = \cos^{-1}\left(\frac{(z_r - z_{p1}) \cdot m \cdot cos(\alpha)}{2 \cdot a}\right)$$

With a being the center distance. This equation for avoiding involute interference can be used as long as $d_{ar} \geq d_{br}$ and for $\alpha = 20°$, also $z_r \geq 35$. We can isolate the topcircle diameter of the ring gear from these equations

$$d_{ar} \geq \frac{d_{br}}{\cos(\tan^{-1}(\tan(\alpha_w)\,(1 - \frac{z_{p2}}{z_r})))}$$

The thickness tip of the tooth of a hardened tooth must be at least 0,4 times the module. Some of this is taken into account by profile shifting but it may also be required to reduce the topcircle diameter in order to have the required thickness. The thickness at the tip is calculated

$$s_a = d_a \cdot \left(\frac{2\pi + 4 \cdot x \cdot \tan(\alpha)}{4 \cdot z} + inv(\alpha) - inv(\alpha_a)\right) \geq s_{a\,min}$$

Where we have the involute function (in radians)

$$inv(\alpha) = \tan(\alpha) - \alpha$$

To ensure this minimum, the thickness is first checked if it is above the minimum and if not, newton-raphsons iterations are used to find the topcircle diameter that will ensure this by finding the solution of

$$\Phi = d_a \cdot \left(\frac{2\pi + 4 \cdot x \cdot \tan(\alpha)}{4 \cdot z} + inv(\alpha) - inv(\alpha_a) \right) - s_{a\,min} = 0$$

$$\frac{\partial \Phi}{\partial d_a} = \frac{2\pi + 4 \cdot x \cdot \tan(\alpha)}{4 \cdot z} + inv(\alpha) - inv(\alpha_a)$$

With the initial guess $d_{a\,0} = d + 2 \cdot m_n$, the following is iterated until the tip diameter has converged.

$$d_{a\,(i+1)} = d_{a\,i} - \frac{\Phi_i}{\left(\frac{\partial \Phi}{\partial d_a}\right)_i}$$

To ensure smooth transmission of power of a gear pair, it is required that the overlap ratio is more than 1,1 and recommended that it is above 1,25. The overlap ratio of a gearset is the average amount of gear tooth pairs in contact such that if it is 1, then one pair carries the load at any time, if it is between 1 and 2, then two pairs carries the load some of the time and one the rest of the time and if it is more than 2, then multiple pairs carries the load at all times. For spur gears, the overlap is calculated

$$\varepsilon_\alpha = \frac{0{,}5 \cdot \left(\sqrt{d_{a1}^2 - d_{b1}^2} + \frac{z_2}{|z_2|} \cdot \sqrt{d_{a2}^2 - d_{b2}^2} \right) - a \cdot \sin(\alpha_w)}{\pi \cdot m \cdot \cos(\alpha)}$$

So a constraint of $\varepsilon_\alpha \geq 1{,}25$ is set for both the gear engagement at the sun and the engagement at the ring.

Geometric constraints specific to our configuration

To keep the costs low, a constraint is set that the module of all gears must be the same.

In DS/ISO 6336-3, it is stated that the strength of a gear against fracture at the foot is reduced if the amount of material underneath a tooth is less than 1,2 times the tooth height and since the tooth height is 2,25 times the module, a constraint is set that the pitch diameter of a planet minus 6 times the module must be larger than or equal to the outer diameter of the needle bearing.

$$d_p - 6 \cdot m \geq d_{need \quad outer}$$

This means that the strength lowering rim thickness factor due to this can at most be 1,75.

To only consider the most compact configurations, constraints are also set for the ring gear including material to hold onto to be less than the outer diameter of the biggest bearing that could be considered.

$$d_r - 15[mm] \geq -130[mm]$$

And a constraint is also set for the swept outer diameter of the large planets. This was initially also set for 130mm, but as it was found that restricting this to 95mm would enable a more beneficial placement in regards to the brakes, the constraint was set to this.

$$d_s + 2 \cdot \left(d_{p1} + m\right) \leq 95\ [mm]$$

Also constrained is that the greatest common denominator of a the amount of teeth in a gear pair must be 1 as this spreads the wear onto all of the teeth by not having any pair of teeth engage more often than others in that gear pair.

Finally, a constraint is set that the pitch diameter of the sun must be more than 14mm. This is set as iterations of FEM analysis concluded that at diameters smaller than this, the torsional stiffness of the sun would not be sufficient to spread the load far enough along the gear tooth that increasing the width would lead to a capable design. Also, increasing the diameter of the sun decreases the loads on all engagements to such an extent that the following was found as a lower limit:

$$d_s > 14$$

Materials

The gears are to be made from 34CrNiMo6 as this is a high strength steel, which is commercially available and that can be hardened. For many iterations, 42CrMo4 was considered as this was recommended by CFT Tandhjulsfabrikken and since limit strength of reference gears is available in TB 20-1 of [1] but late in the project, 34CrNiMo4 was chosen as it has similar strengths as 42CrMo4 but that the Nickel content makes for better thermal stability and hardenability according to Bodycote.

The strengths for 42CrMo4 in a gas-nitrided state is assumed. These are assumed to be conservative as they are valid for hardnesses of 48-57 HRC and the nitriding will usually result in hardnesses of 60-62 HRC according to Bodycote.

$$\sigma_{Flim} = 430\ [MPa]$$

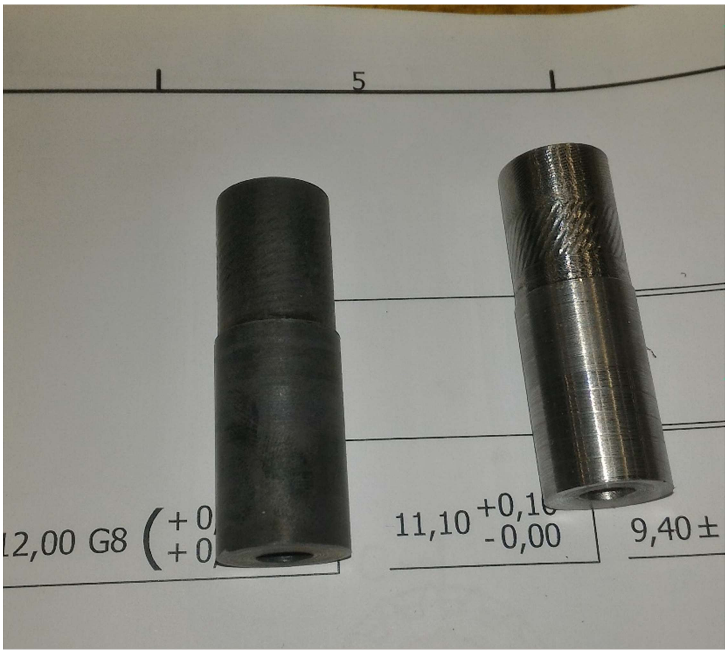
$$\sigma_{Hlim} = 1215\ [MPa]$$

Machining of the gear teeth is expected to produce a tooth quality of no better than 8 according to DS/ISO 1328-1 and the geometry of these gears prevents grinding of the teeth so in order to limit distortion due to release of internal stresses in the hardening process, it is chosen that the manufacturing process should be:

1) Rough machining of blanks, 1mm from final dimensions
2) Stress releasing heat treatment
3) Final machining of blanks
4) Gear tooth machining
5) Gas nitriding, 20 hours, expected hardening depth of 0,18-0,35mm.
6) Grinding of bearing surfaces

The stress releasing heat treatment consists of heating the blanks to 550-580 degrees for 2 hours and letting them cool slowly.
When heated to 560 degrees[2] for 2,5 hours in atmospheric air in a hardening oven at the university, mill scales formed but were found to be very superficial. On the sun gear blanks, no deviations was measurable with a caliper after stress releasing heat treatment.

[2] Overshot by 4 degrees for a short time during warmup. +/- 1 degree otherwise



Tooth root bending stress

The stress at the tooth root will be calculated mostly according to ISO 6336-3:2006 from the form

$$\sigma_F = \frac{F_t}{b \cdot m_n} \cdot Y_F \cdot Y_S \cdot Y_B \cdot Y_\varepsilon \cdot K_A \cdot K_V \cdot K_{F\alpha} \cdot K_{F\beta}$$

Where

- Y_F is the form factor, which takes into account how the force is applied on the tooth.
- Y_S is the stress correction factor, which takes the shape of the tooth root into account to calculate the maximum stress.
- Y_B is the rim thickness factor, which for gears with thin rims adjusts the root stress.
- Y_ε is a factor that takes into account that the load is partially shared between multiple teeth. If $\varepsilon <$ 2, then one set of teeth will at some time be the only set in contact but since this will be when the contact is near the middle of the tooth rather than at any of the extremes, this factor still reduces the stress since Y_F adjusted the bending stress corresponding to a load applied at the tip of the tooth.
- K_A is the application factor, which takes uncertainties and oscillations in applied load into account.
- K_V is the dynamic factor, which takes internal vibrations into account. Since these gears are to be very light, this factor is found to contribute very little but it is included nonetheless.
- $K_{F\alpha}$ is the transverse load factor and takes into account that the transverse load is not uniform.
- $K_{F\beta}$ is the face load factor which takes into account that the load is not uniform across the width of the tooth face. This is due to the twist of the gears, manufacturing misalignments, bearing clearances and the elastic deflections of the gear shafts, planet carrier and housing. Due to our gears having a high with in relation to their diameter, the methods in ISO 6336-1 has not been able to provide a meaningful result of this factor and thus FEM analysis is used.

The tooth root stress limit is calculated

$$\sigma_{FG} = \sigma_{Flim} \cdot Y_{ST} \cdot Y_{NT} \cdot Y_M \cdot Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X$$

- Y_{ST} is the stress correction factor of the reference gears. For data according to ISO 6336-5, $Y_{ST} = 2$.
- Y_{NT} is the lifetime factor, which takes into account that the allowable stress is higher if the load cycles is less than $3 \cdot 10^6$. For multiple different loadings at different amount of cycles, Miner's rule could be used to calculate the life of the gears as described in ISO 6336-6 but since this requires reliable measurements in the condition that it will be used, this is omitted.
- Y_M is the mean stress influence factor which reduces the allowable stress due to a shift in the direction of the applied torque.
- $Y_{\delta relT}$ is the relative notch sensitivity factor and takes the notch sensitivity of the material into account. $Y_{\delta relT} \approx 1$ but it is included as it seems to have 0,5-2% of influence in the direction of reducing allowable stress.
- Y_{RrelT} is the relative surface factor. Since the nitride steel isn't as sensitive to roughnesses as hardened steel and the roughness isn't actually known beforehand, this factor is assumed to be unity.
- Y_X is the size factor which is unity for $m_n \leq 5$.

With these, the tooth width can be isolated as

$$b = \frac{\frac{F_t}{m_n} \cdot Y_F \cdot Y_S \cdot Y_B \cdot Y_\varepsilon \cdot K_A \cdot K_V \cdot K_{F\alpha} \cdot K_{F\beta}}{\sigma_{Flim} \cdot Y_{ST} \cdot Y_{NT} \cdot Y_M \cdot Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X} \cdot SF_{min}$$

Implementation of the calculations of the ISO 6336-3 standard has only been possible for external gears, so the strength of the ring gear is assumed to be at least as much as the smaller planet. This assumption is considered conservative as the wear and damage will be spread onto a larger number of teeth and since the teeth of the ring gear are wider at the base than those of the planet due to the ring being internally geared.

Pitting

Implementation of the calculations in the ISO 6336-2 standard has not been successful in providing meaningful results.

Matlab implementation and optimization

To choose an optimum design of a compound planetary gearbox, it is sought to minimize the combined kinetic energy which is the sum of the energy from longitudinal acceleration of the complete mass, the rotational acceleration of the planet carrier, planet shafts and planets around the wheel axis, the rotational acceleration of the planets around the planet shafts and the rotational acceleration of the sun gear around the wheel axis.
To describe these masses and inertias as a function of gear ratios and strength constraints in a way that can be differentiated and optimized by a classical approach is difficult so instead, the massive computing power of a contemporary laptop to do an exhaustive search optimization.

For-loops are used to run through all possible combinations of amount of gear teeth less than 120 teeth and modules between 0,8mm and 1,2mm.

To limit the computational requirement of the optimization, if-statements are used to only continue calculations on a certain configuration if the gear ratio is within the wanted range and if the geometrical constraints are fulfilled.

FEM face load factor

A method in [5] is to use FEM to calculate the contact pressure and then multiply by the element height to get the load per unit width in one element and sum it up across the tooth height and by that finding the equivalent line load across the tooth. Then by comparing the maximum against the average, the face load factor is calculated. This makes it possible to include all deformations of the carrier and the gears.

The face load factor is then defined as the maximum load per unit length divided by the average load per unit length

$$K_{H\beta} = \frac{\left(\frac{F_m}{b}\right)_{max}}{\left(\frac{F_m}{b}\right)}$$

Results

Through 10 iterations, the method of analyzing the face loads through FEM and using it in the matlab code was refined and the following configuration was found to be the optimum:

$$i = 11{,}0118, z_s = 15, z_r = 69\ z_{p1} = 37, z_{p2} = 17$$

$$m_s = m_{p1} = m_r = m_{p2} = 1\ [mm]$$

$$n_{planets} = 3$$

$$b_s = b_{p1} = 7{,}7\ [mm], b_r = b_{p2} = 14{,}5\ [mm]$$

$$calculated\ mass = 0{,}968[kg]$$

$$calculated\ equivalent\ mass = 1{,}152[kg]$$

The manufacturing misalignment will be calculated into a deflection at the ends of the planet shaft by assuming small angles so $\tan(\theta) \approx \theta = \frac{f_{ma}}{b}$. If we assume the value of f_{ma} with tooth quality 8 to be the value at a $\pm 6\sigma$ variance, and allow for 1,35% of the planets to be above this value, then we can half it as we would then be at $\pm 3\sigma$ which only 2,7% is outside of but since it is only a problem if it is in the direction that adds to the total deflection, we will have more safety.

$$\delta_{ma} = \frac{f_{ma}}{b} \cdot l$$

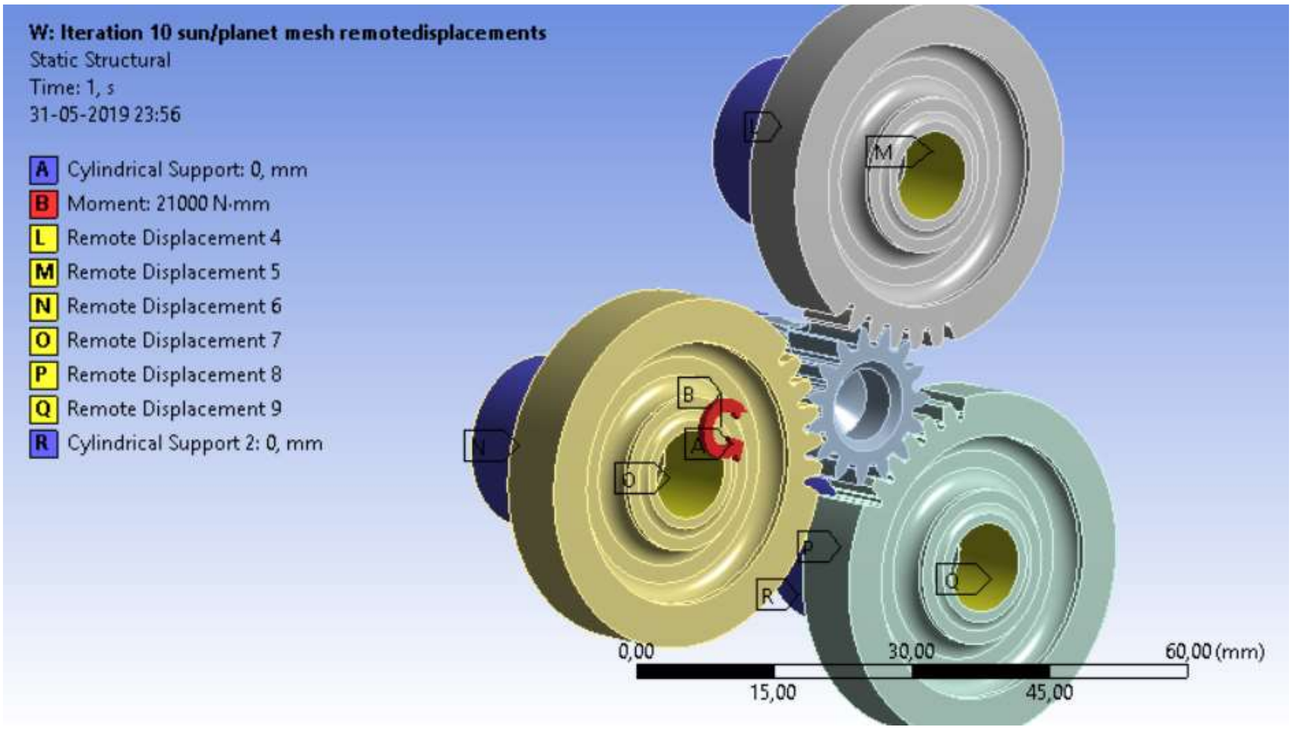
With l being the length to the end that the deviation is applied, with the deviation assumed zero at the center of the gear tooth. The deflections are applied in the direction that gives the worst outcome.

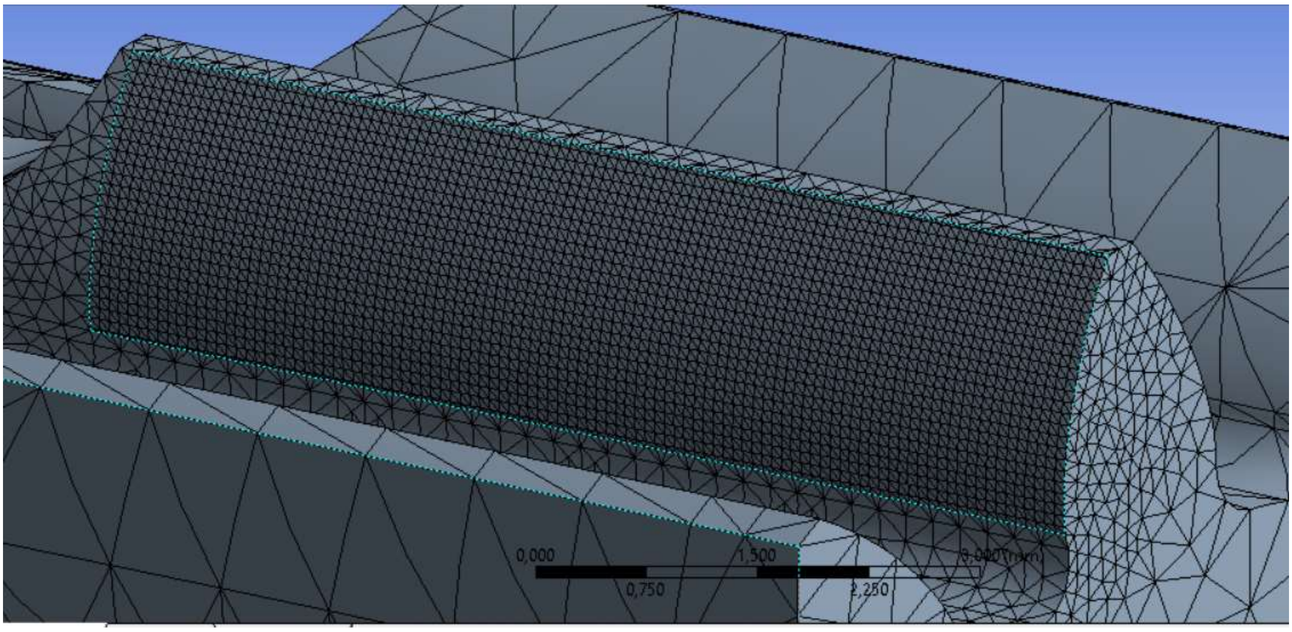
For the carrier deflections to be found, an FEM analysis is made where the loads are applied directly at the planet gears which are modeled as cylinders. For the sun mesh, the deflections are

$$\delta_1 = (\delta_{ca1} + \delta_{be1} + \delta_{ma}\ ) \cdot 0{,}85 = \left(\begin{Bmatrix} 9{,}216 \\ -0{,}242 \end{Bmatrix} + \begin{Bmatrix} 7{,}936 \\ -2{,}888 \end{Bmatrix} + \begin{Bmatrix} -5{,}602 \\ 1{,}023 \end{Bmatrix}\right) \cdot 0{,}85 = \begin{Bmatrix} 9{,}817 \\ -1{,}792 \end{Bmatrix}\ \mu m$$

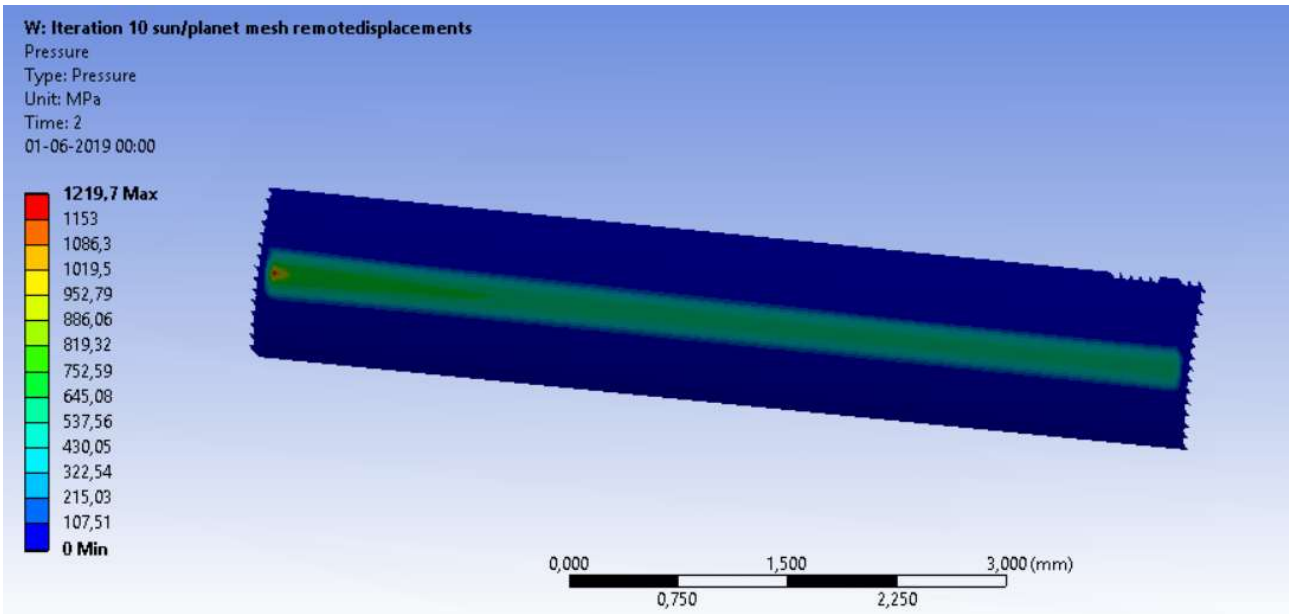
$$\delta_2 = (\delta_{ca2} + \delta_{be2} + \delta_{ma2}) \cdot 0{,}85 = \left(\begin{Bmatrix} 18{,}549 \\ 0{,}947 \end{Bmatrix} + \begin{Bmatrix} 8{,}329 \\ 1{,}393 \end{Bmatrix} + \begin{Bmatrix} 29{,}313 \\ -2{,}552 \end{Bmatrix}\right) \cdot 0{,}85 = \begin{Bmatrix} 47{,}763 \\ -0{,}180 \end{Bmatrix} \mu m$$

These deflections are entered as remote displacements into an FEM analysis of the mesh between the sun and the planets in which one of the gear tooth are in contact at the pitch cylinder of both gears and this tooth is meshed with face mapping with 100 elements in the height.

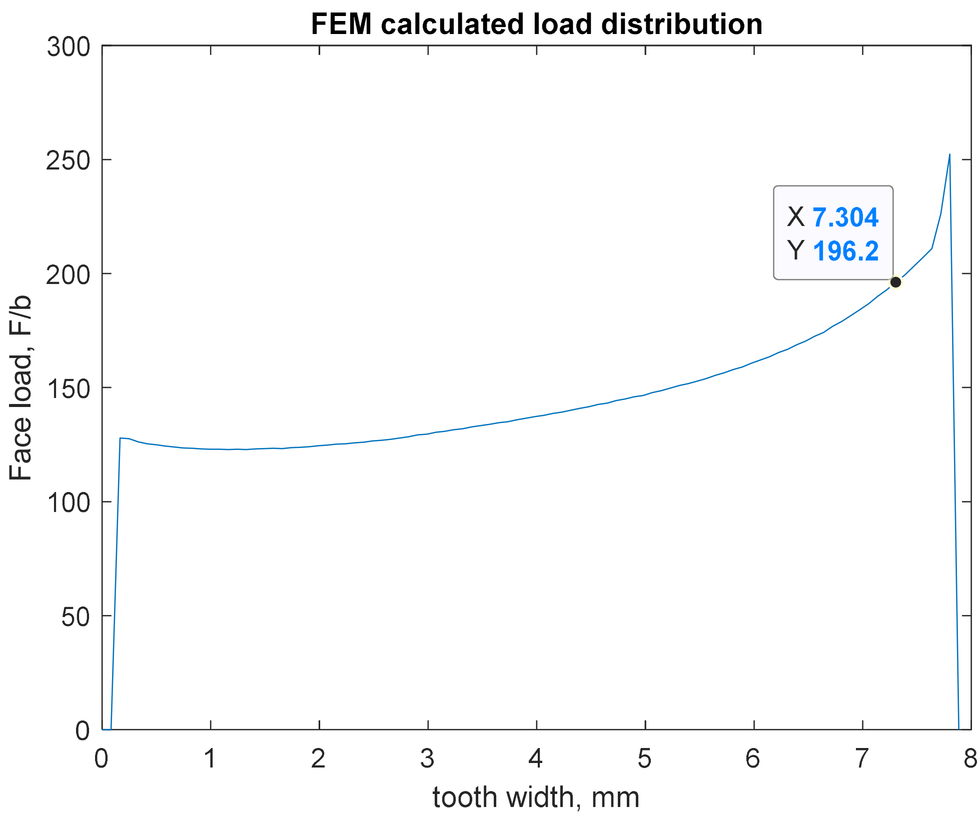




The pressure is exported as a text file and the equivalent line load is calculated from the nodal pressures and the element heights to find the FEM face load factor.



The outermost 0,5mm of the contact are then removed due to chamfers being made onto the edges of the gears.



The face load factor becomes 1,2767 when the peak load is reduced due to chamfer. The load is skewed towards the motor end of the sun mesh.

For the ring mesh, the deflections are

$$\delta_1 = (\delta_{ca1} + \delta_{be1} + \delta_{ma1}) \cdot 0,85 = \left(\begin{Bmatrix} 9,216 \\ -0,242 \end{Bmatrix} + \begin{Bmatrix} 7,936 \\ -2,888 \end{Bmatrix} + \begin{Bmatrix} -14,016 \\ 2,558 \end{Bmatrix}\right) \cdot 0,85 = \begin{Bmatrix} 2,665 \\ -0,4865 \end{Bmatrix} \mu m$$

$$\delta_2 = (\delta_{ca2} + \delta_{be2} + \delta_{ma2}) \cdot 0,85 = \left(\begin{Bmatrix} 18,549 \\ 0,947 \end{Bmatrix} + \begin{Bmatrix} 8,329 \\ 1,393 \end{Bmatrix} + \begin{Bmatrix} 5,8947 \\ -0,5132 \end{Bmatrix}\right) \cdot 0,85 = \begin{Bmatrix} 27,857 \\ 1,553 \end{Bmatrix} \mu m$$

And the face load factor calculates to 1,881 with the peak removed due to chamfered ends. This results in the factors of safety

$$SF_{sun} = 1,48$$

$$SF_{p1} = 1,50$$

$$SF_{p2} = 1,21$$

With the profile shift coefficients $x_s = 0,3$, $x_{p1} = -x_s$, $x_{p2} = 0,2$ and $x_r = -x_{p2}$

References

[1] Wittel, Muhns, Jannash, Voßiek (2013), "Roloff/Matek Machinenelemente + Tabellenbuch" 21st edition. Springer Vieweg. ISBN: 978-3-658-02326-3

[2] Dudley, D.W (1962), "Gear handbook". 1st edition. McGraw-Hill

[3] Petru A. Simionescu, "A Unified Approach to the Assembly Condition of Epicyclic Gears". Journal of Mechanical Design, sept. 1998.

[4] QTCgears catalog technical section, "Elements of metric gear technology". http://qtcgears.com/tools/catalogs/PDF_Q420/Tech.pdf, Accessed 31/05-2019

[5] Roda-Casanova et al. "Determination of the ISO face load factor in spur gear drives by the finite element modeling of gears and shafts". Mechanism and Machine Theory, March 2013.